

46,XY Sex Reversal 11 (SRXY11): A Comprehensive Disease Characteristics Report

Disease: 46,XY Sex Reversal 11 (SRXY11) **Causal gene:** *DHX37* (DEAH-box RNA helicase 37) **OMIM (phenotype):** #273250 (46,XY sex reversal 11) **Category:** Mendelian, autosomal dominant disorder/difference of sex development (DSD) **Suggested disease ontology mapping:** MONDO — 46,XY partial/complete gonadal dysgenesis spectrum; the *DHX37*-specific entry corresponds to OMIM #273250.

Evidence source key: [H] human clinical/genetic · [M] model organism · [V] in vitro/functional · [C] computational/in silico · [R] review.

Summary

46,XY Sex Reversal 11 (SRXY11) is a rare Mendelian disorder/difference of sex development (DSD) caused by heterozygous, mostly recurrent missense variants in *DHX37*, a DEAH-box RNA helicase essential for biogenesis of the small (40S) ribosomal subunit. Affected individuals have a 46,XY karyotype but fail to complete testis determination or undergo testis regression, producing a clinical spectrum that runs from phenotypic females with complete gonadal dysgenesis (Swyer syndrome), through partial gonadal dysgenesis, to testicular regression syndrome (TRS)/anorchia and, at the mildest pole, males with modest testicular underdevelopment and gynecomastia. Because a ubiquitously required housekeeping factor produces a phenotype restricted almost entirely to the developing gonad, *DHX37*-related DSD has been proposed as a new human **ribosomopathy** — a class of disease in which broadly expressed ribosome-biogenesis factors nonetheless cause tissue-specific pathology.

The molecular link between *DHX37* loss and gonadal failure is coming into focus from mouse work: *DHX37* safeguards nucleolar integrity and PI3K-AKT survival signaling and suppresses p53-driven apoptosis, so its deficiency triggers pro-apoptotic RNA splicing and death of the fetal supporting (Sertoli) cell lineage, aborting or reversing testis formation. In humans, *DHX37*

protein is expressed principally in germ cells and Leydig cells, and only rarely in Sertoli cells, so the exact cell-autonomous versus non-cell-autonomous route to Sertoli-cell failure remains partially inferred. Two recurrent variants — **p.Arg308Gln** and **p.Arg674Trp** — account for a large share of cases and are especially associated with embryonic testicular regression syndrome (ETRS).

Clinically, the disorder is diagnosed by the combination of a 46,XY karyotype discordant with gonadal/genital phenotype, hypergonadotropic hypogonadism (elevated FSH/LH, low sex steroids), imaging showing streak or absent gonads (with or without Müllerian structures), and molecular confirmation by whole-exome/genome sequencing, for which DHX37 should now be part of DSD gene panels. Management is supportive rather than curative: sex-steroid hormone replacement, prophylactic gonadectomy of dysgenetic Y-bearing gonads (which carry a ~15–23% **germ-cell tumor risk**), fertility and psychosocial counseling, and genetic counseling. Importantly, biallelic and de novo heterozygous DHX37 variants cause a **distinct, allelic neurodevelopmental syndrome (NEDBAVC, OMIM 618731)**, so genotype must be interpreted with the full clinical picture.

Key Findings

Finding 1 — DHX37 heterozygous missense variants are a frequent, autosomal-dominant cause of 46,XY DSD (SRXY11) [H]

In a cohort of 145 individuals with 46,XY DSD of previously unknown etiology, 13 children carried heterozygous missense pathogenic variants in *DHX37*, and rare/novel DHX37 missense variants were enriched in cases versus controls with high statistical significance ($P = 5.8 \times 10^{-10}$). The gene encodes an RNA helicase "essential for ribosome biogenesis," and the pathogenic variants establish an **autosomal dominant** form of 46,XY DSD encompassing both gonadal dysgenesis and testicular regression syndrome (TRS). The authors concluded these conditions "are part of a clinical spectrum" rather than distinct entities.

"Thirteen children carried heterozygous missense pathogenic variants involving the RNA helicase DHX37, which is essential for ribosome biogenesis." — [PMID: 31337883](#)

"DHX37 pathogenic variants are a new cause of an autosomal dominant form of 46,XY DSD, including gonadal dysgenesis and TRS, showing that these conditions are part of a clinical spectrum." — [PMID: 31337883](#)

This is the foundational human-clinical evidence identifying DHX37 as the SRXY11 gene and framing the phenotype as a spectrum. HGNC: DHX37. UniProt: Q8IY37.

Finding 2 — DHX37 maintains supporting-cell (Sertoli) survival through nucleolar integrity and PI3K-AKT, suppressing p53 apoptosis [M][V]

Multi-omics analysis of cell-specific *Dhx37* knockout mice (RIP-seq plus RNAi-RNA-seq) demonstrated that *Dhx37* "safeguards nucleolar integrity and PI3K-AKT signaling, suppresses p53-driven apoptosis, and its loss triggers pro-apoptotic splicing" and Sertoli-cell death, impairing testis development. This is the strongest mechanistic account currently available and links DHX37 loss-of-function to the cellular event (supporting-cell apoptosis) that most plausibly explains failed/reversed testis determination.

"Dhx37 safeguards nucleolar integrity and PI3K-AKT signaling, suppresses p53-driven apoptosis, and its loss triggers pro-apoptotic splicing" — [PMID: 41535247](#)

Relevant GO/pathway terms: ribosome biogenesis (GO:0042254), rRNA processing (GO:0006364), nucleolus (GO:0005730), PI3K-AKT signaling, intrinsic apoptotic signaling by p53 class mediator (GO:0072332), regulation of RNA splicing (GO:0043484).

Finding 3 — Allelic heterogeneity: heterozygous → non-syndromic DSD; biallelic/de novo → syndromic NEDBAVC (OMIM 618731) [R][H]

DHX37 shows a clean genotype-driven dichotomy. Recurrent **heterozygous** missense variants — affecting highly conserved residues in the helicase domains and predicted deleterious — cause **non-syndromic** 46,XY gonadal dysgenesis, TRS, or anorchia. In contrast, **compound heterozygous and de novo heterozygous** DHX37 missense variants cause a complex congenital syndrome, **NEDBAVC** (neurodevelopmental disorder with brain anomalies and with/without vertebral or cardiac anomalies; OMIM 618731) featuring microcephaly, global developmental delay, seizures, facial dysmorphism, and kidney/cardiac anomalies.

"compound heterozygous as well as de novo heterozygous missense variants in DHX37 are also associated with a complex congenital developmental syndrome (NEDBAVC, neurodevelopmental disorder with brain anomalies and with or without vertebral or cardiac anomalies; OMIM 618731), consisting of microcephaly, global developmental delay, seizures, facial dysmorphism, and kidney and cardiac anomalies" — [PMID: 35835064](#)

"All affected children have non-syndromic forms of disorders/differences of sex development (DSD)." — [PMID: 35835064](#)

This distinction is clinically critical for variant interpretation and counseling.

Finding 4 — Dysgenetic Y-bearing gonads carry a high germ-cell tumor risk, warranting gonadectomy [H]

In a series of 292 phenotypic-female DSD patients harboring Y-chromosome material, the overall germ-cell tumor (GCT) risk was **15.4%**, and **46,XY pure gonadal dysgenesis carried the highest risk (~23.3%)**. Tumors — gonadoblastoma and dysgerminoma/seminoma — arose predominantly during adolescence (median 17–18 years). Notably, no tumor was found in five testicular-regression patients, contrasting the high dysgenesis risk with the negligible regression risk. These data support **prophylactic gonadectomy** of dysgenetic gonads.

"The overall GCTs risk was 15.41% and 46, XY pure gonadal dysgenesis (46, XY PGD) carried the highest risk up to 23.33%" — [PMID: 27862157](#)

"no tumour was found in five testis regression patients" — [PMID: 27862157](#)

A familial Swyer-syndrome report additionally suggests familial cases may carry higher tumor risk than sporadic ones (66.6% vs. 15–45%) ([PMID: 38337479](#)).

Finding 5 — Marked phenotypic heterogeneity from complete gonadal dysgenesis to TRS/anorchia, with variable expressivity [H]

DHX37 variants generate a phenotypic continuum: 46,XY complete gonadal dysgenesis (female external genitalia, Müllerian remnants), partial gonadal dysgenesis, testicular regression syndrome/anorchia (absent testes with variable male genitalia), and milder testicular underdevelopment with gynecomastia. Variants cluster in conserved helicase domains (e.g., RecA1); reported residues include p.Arg334Trp, p.Arg390His, p.Thr477His, and p.Gly478Arg. A

striking example of variable expressivity/incomplete penetrance: a boy with TRS carried a homozygous p.T477H variant while his **fertile father**, carrying the same variant, had only unilateral testicular regression with otherwise typical male genital development.

"Missense variants in the RNA-helicase DHX37 are associated with either 46,XY gonadal dysgenesis or 46,XY testicular regression syndrome (TRS)." — [PMID: 34293745](#)

"a homozygous p.T477H variant was identified in a boy with TRS. His fertile father had unilateral testicular regression with typical male genital development." — [PMID: 34293745](#)

"manifestations ranging from complete gonadal dysgenesis to mild testicular underdevelopment with gynecomastia" — [PMID: 42510869](#)

Finding 6 — Diagnosis rests on 46,XY karyotype discordant with gonad/genitalia, hypergonadotropic hypogonadism, and molecular (WES/WGS) confirmation [H]

Complete gonadal dysgenesis (Swyer syndrome) typically presents in phenotypic females with a 46,XY karyotype, primary amenorrhea/delayed puberty, a hypoplastic uterus and streak gonads on imaging, and **hypergonadotropic hypogonadism** (elevated FSH/LH, low estradiol/testosterone) on hormonal assay. DHX37 is identified by whole-exome/trio sequencing once karyotype excludes sex-chromosome DSD; the gene is now recommended for inclusion in DSD gene panels and genome-wide sequencing.

"confirmed by the hormonal assay that showed hypergonadotropic-hyp[ogonadism]" — [PMID: 37497464](#)

"Genome-wide sequencing should be prioritized in VSC/DSD diagnostics, consistent with current best practices, to improve diagnostic yield" — [PMID: 41466375](#)

Finding 7 — Two recurrent variants (p.Arg308Gln, p.Arg674Trp) dominate and are enriched vs. gnomAD, with a specific ETRS association; DHX37 is expressed in germ and Leydig cells [H][V]

In 87 patients with 46,XY DSD, da Silva et al. (2019) identified pathogenic/likely-pathogenic heterozygous DHX37 missense variants in 5 families (11 patients) and 6 sporadic cases; two recurrent variants dominated — **p.Arg308Gln** (two families, three sporadic cases) and

p.Arg674Trp (two families, two sporadic cases). Rare, predicted-deleterious DHX37 variants occurred in **14%** of the cohort versus **0.4%** in gnomAD (**P < 0.001**), and were specifically associated with embryonic testicular regression syndrome (ETRS) in **7/14 index cases (50%)**. Immunohistochemistry localized DHX37 mainly to germ cells (at various maturation stages) and Leydig cells, and only rarely to Sertoli cells.

"Two variants were recurrent: p.Arg308Gln (in two families and in three sporadic cases) and p.Arg674Trp (in two families and in two sporadic cases)." — [PMID: 31287541](#)

"The frequency of rare, predicted-to-be-deleterious DHX37 variants in this cohort (14%) is significantly higher than that observed in the Genome Aggregation Database (0.4%; P < 0.001)." — [PMID: 31287541](#)

"DHX37 is mainly expressed in germ cells at different stages of testis maturation, in Leydig cells, and rarely in Sertoli cells" — [PMID: 31287541](#)

"The variants were specifically associated with ETRS (7/14 index cases; 50%)." — [PMID: 31287541](#)

Finding 8 — The central mechanism — how a ubiquitous ribosome-biogenesis factor produces tissue-specific gonadal failure — remains unresolved [R]

DHX37 is a housekeeping DEAH-box helicase required for 40S ribosomal subunit biogenesis in every cell, yet heterozygous missense variants produce a phenotype almost entirely restricted to gonad/testis determination — a paradox shared with other ribosomopathies. Multiple primary reports and reviews explicitly state that the pathogenic mechanism is unknown.

"Similar to all other known ribosomopathies, the mechanism of pathogenesis is unknown." — [PMID: 34293745](#)

"DHX37 is required for ribosome biogenesis, and this subgroup of XY DSD is a new human ribosomopathy." — [PMID: 34293745](#)

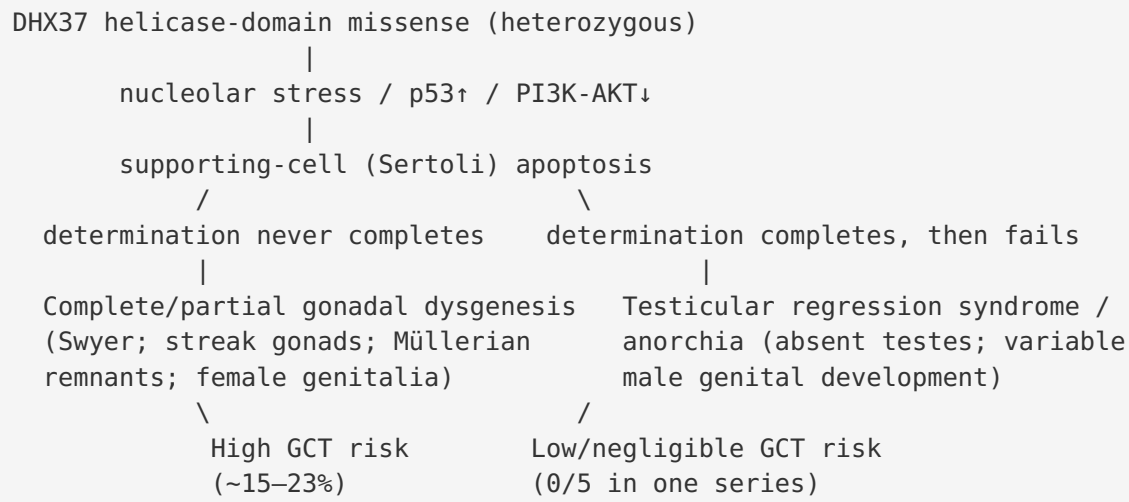
No zebrafish or invertebrate DHX37 sex-development model, and no humanized knock-in of the recurrent p.Arg308Gln/p.Arg674Trp alleles, has yet been reported; available models are a supporting-cell conditional-knockout mouse and yeast Dhr1 structural/functional studies.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous missense variant in DHX37** (most often at recurrent hotspots p.Arg308Gln or p.Arg674Trp, in conserved helicase RecA domains) *leads to* a partial loss of DHX37 RNA-helicase activity. (*Demonstrated genetically; the loss-of-function nature at the protein level is partly inferred from conservation, in silico prediction, and yeast Dhr1 structural work — [PMID: 31188444].*)
2. Impaired DHX37 helicase function *results in* defective displacement of the U3 snoRNA from pre-rRNA and impaired **40S small ribosomal subunit biogenesis / nucleolar homeostasis**. (*Mechanism established for yeast Dhr1; extrapolated to human DHX37.*)
3. Disturbed nucleolar integrity *leads to* **nucleolar stress**, which *results in* stabilization/activation of **p53** and reduced **PI3K-AKT survival signaling**. (*Demonstrated in Dhx37-knockout mouse multi-omics — [PMID: 41535247].*)
4. p53 activation plus loss of AKT survival signaling *triggers* **pro-apoptotic RNA splicing and apoptosis of the fetal supporting (Sertoli) cell lineage**. (*Demonstrated in mouse; human Sertoli involvement is partly inferred because human DHX37 protein is expressed mainly in germ and Leydig cells and only rarely in Sertoli cells — [PMID: 31287541].*)
5. Loss/failure of Sertoli-supporting cells during the narrow window of fetal sex determination *results in* **failed testis determination (gonadal dysgenesis)** or, if determination initially succeeds, **testis regression (TRS/anorchia)**. (*Branch point — see below.*)
6. Absent or dysgenetic testis *leads to* deficient anti-Müllerian hormone and androgen output, which *results in* **incomplete/absent virilization, persistence of Müllerian structures (in dysgenesis), and hypergonadotropic hypogonadism** at expected puberty. (*Human clinical — [PMID: 37497464].*)
7. Retained dysgenetic Y-bearing gonadal tissue *creates a* **germ-cell tumor predisposition (gonadoblastoma → dysgerminoma/seminoma)**. (*Human clinical — [PMID: 27862157].*)

Branch point



Genotype–phenotype and dose logic

Genotype	Phenotype	Syndromic?	OMIM
Heterozygous missense (helicase domains; hotspots R308Q, R674W)	46,XY gonadal dysgenesis ↔ TRS/ anorchia spectrum	No	#273250 (SRXY11)
Compound heterozygous / de novo heterozygous missense	Microcephaly, DD, seizures, dysmorphism, vertebral/cardiac/ kidney anomalies	Yes (NEDBAVC)	618731

The recurrence of specific residues and their autosomal-dominant behavior suggest the DSD-causing alleles act through a **specific (possibly dominant-negative or hypomorphic gain-of-toxicity) mechanism** rather than simple haploinsufficiency — consistent with the observation that a different mutational configuration (biallelic/de novo) produces an entirely different, neurodevelopmental disease. This remains a hypothesis; direct allele-specific functional proof is lacking.

Why a housekeeping factor hits the gonad selectively (open question)

The defining unresolved question (Finding 8) is the **tissue-specificity paradox**. Proposed but unproven explanations, by analogy to other ribosomopathies, include: (i) heightened dependence of the rapidly proliferating fetal gonadal-somatic lineage on ribosome flux during

the brief sex-determination window; (ii) a low p53 threshold in supporting cells; and (iii) selective translational requirements for pro-testis regulators (e.g., the SRY/SOX9 pathway). None has been experimentally demonstrated for DHX37.

Section-by-Section Details

1. Disease Information

- **Overview:** SRXY11 is a monogenic 46,XY DSD in which testis determination fails or reverses despite a Y chromosome, spanning Swyer syndrome (complete gonadal dysgenesis) to testicular regression/anorchia to mild male undervirilization.
- **Identifiers:** OMIM #273250 (46,XY sex reversal 11); allelic OMIM 618731 (NEDBAVC). Gene: *DHX37* (HGNC:16192; NCBI Gene 57647; UniProt Q8IY37). Orphanet/ICD map to broader "46,XY complete/partial gonadal dysgenesis" and "swyer syndrome" categories; MeSH: Gonadal Dysgenesis, 46,XY; Disorders of Sex Development. MONDO: 46,XY gonadal dysgenesis spectrum.
- **Synonyms/alternatives:** DHX37-related 46,XY DSD; 46,XY gonadal dysgenesis with DHX37; 46,XY testicular regression syndrome (DHX37); "new ribosomopathy of sex development."
- **Data source type:** Predominantly aggregated case-series/cohort literature and gene-level resources (OMIM), not EHR-derived.

2. Etiology

- **Causal factor:** Monogenic — heterozygous missense variants in *DHX37* (Findings 1, 7). No environmental or infectious cause.
- **Genetic risk:** The DHX37 variant itself is causal (autosomal dominant); rare deleterious DHX37 variants enriched ~35-fold vs gnomAD (14% vs 0.4%; [PMID: 31287541]). Modifier genes are implied by incomplete penetrance but unidentified.
- **Protective factors:** None established (no known protective alleles or exposures).
- **Gene–environment interactions:** None documented; considered a purely genetic condition.

3. Phenotypes

Phenotype	Type	Onset	Frequency	HPO suggestion
46,XY sex reversal / gonadal dysgenesis	Physical/ clinical	Fetal (manifest at birth or puberty)	Core feature	HP:0000133
Streak gonads	Clinical sign	Fetal/ congenital	Common in CGD	HP:0000133
Primary amenorrhea / delayed puberty	Clinical sign	Adolescence	Common in Swyer	HP:0000783
Hypergonadotropic hypogonadism	Lab abnormality	Puberty	Characteristic	HP:0000815
Testicular regression / anorchia	Physical	Fetal	ETRS subset (~50% of variant carriers, [PMID:31287541])	HP:0000795
Gynecomastia (mild pole)	Physical	Puberty	Mild presentations	HP:0000771
Müllerian remnants (uterus)	Physical	Congenital	Dysgenesis end	—
Gonadoblastoma/ dysgerminoma predisposition	Neoplasm	Adolescence	~15–23% in dysgenesis	HP:0100728

Severity is highly variable (Finding 5); the primary gonadal defect is stable/non-progressive but its consequences (pubertal failure, tumor risk) evolve. Quality-of-life impacts include infertility, need for lifelong hormone therapy, psychosocial burden of atypical sex development, and cancer surveillance/gonadectomy; disease-specific QOL instruments have not been reported for this ultra-rare condition.

4. Genetic/Molecular Information

- **Causal gene:** *DHX37* (DEAH-box helicase 37); encodes an ATP-dependent RNA helicase required for 40S biogenesis.

- **Variant class:** Missense predominates, in conserved helicase (RecA) domains. Recurrent hotspots: **p.Arg308Gln**, **p.Arg674Trp** ([PMID: 31287541]). Other reported residues: p.Arg334Trp, p.Arg390His, p.Thr477His, p.Gly478Arg ([PMID: 34293745]).
- **Classification:** Pathogenic/likely-pathogenic per ACMG for hotspots; some variants remain VUS requiring functional/segregation support ([PMID: 42057034]).
- **Allele frequency:** Rare/near-absent in gnomAD (aggregate deleterious ~0.4%).
- **Origin:** Germline; both inherited (dominant) and de novo reported.
- **Functional consequence:** Loss/alteration of helicase function; a dominant-negative or hypomorphic mechanism is hypothesized but unproven.
- **Modifier genes/epigenetics/chromosomal:** Modifiers unidentified; no epigenetic mechanism established; the causal lesion is a point mutation, not a large chromosomal rearrangement (distinguishing it from 45,X/46,XY mosaicism).

5. Environmental Information

Not applicable — no environmental, lifestyle, or infectious contributors are known; SRXY11 is monogenic.

6. Mechanism / Pathophysiology

See the ordered causal chain and branch diagram above. Molecular pathway: ribosome biogenesis (40S/SSU maturation, U3 snoRNA displacement) → nucleolar stress → p53 activation + reduced PI3K-AKT → pro-apoptotic splicing → supporting-cell apoptosis. GO terms: GO:0042254, GO:0030490, GO:0005730, GO:0072332, GO:0006915, GO:0007530, GO:0008584. CL terms: Sertoli/supporting cell (CL:0000216), Leydig cell (CL:0000178), male germ cell (CL:0000015).

7. Anatomical Structures Affected

- **Organ/system:** Gonad/testis (primary); endocrine and reproductive systems; internal (uterus/Müllerian remnants) and external genitalia (secondary). UBERON: gonad (UBERON:0000991), testis (UBERON:0000473), uterus (UBERON:0000995).
- **Tissue/cell:** Gonadal somatic supporting lineage (Sertoli), Leydig cells, germ cells.
- **Subcellular:** Nucleolus (GO:0005730), cytosolic small ribosomal subunit (GO:0022627).
- **Lateralization:** Usually bilateral (dysgenesis); unilateral regression documented (variable, [PMID: 34293745]).

8. Temporal Development

- **Onset:** Congenital/fetal at the level of gonadal determination; clinically silent until puberty in complete dysgenesis (primary amenorrhea), or noted at birth when genitalia are atypical or testes absent.
- **Course:** Underlying defect fixed/non-progressive; consequences evolve; germ-cell tumors cluster in adolescence (median 17–18 y; [PMID: 27862157]).
- **Critical period:** Fetal sex-determination window (the window of intervention for any future mechanistic therapy is developmental and effectively closed postnatally).

9. Inheritance and Population

- **Inheritance:** Autosomal dominant (SRXY11) with incomplete penetrance and variable expressivity ([PMID: 31337883], [PMID: 34293745]); allelic recessive/de novo → NEDBAVC ([PMID: 35835064]).
- **Epidemiology:** DHX37-specific prevalence/incidence not quantified; identified in enriched DSD cohorts (e.g., 14% of one 87-patient cohort; 13/145 in another). Reported across French, Brazilian, Japanese, Iranian, Chinese, and Polish cohorts.
- **Sex ratio:** Affects 46,XY individuals; presentation ranges phenotypic female → male.
- **Founder/consanguinity:** No founder effect established; homozygous cases reported in consanguineous contexts ([PMID: 34293745]).

10. Diagnostics

- **Cytogenetics:** Karyotype/FISH confirming 46,XY (exclude 45,X/46,XY mosaicism — [PMID: 31883875], [PMID: 32057790]).
- **Endocrine:** Hypergonadotropic hypogonadism (↑ FSH/LH, ↓ estradiol/testosterone); AMH/inhibin B to gauge functional gonadal tissue.
- **Imaging:** Pelvic ultrasound/MRI for uterus/Müllerian structures and streak vs absent gonads.
- **Molecular:** WES/WGS or trio sequencing; DHX37 on DSD panels ([PMID: 41466375], [PMID: 42365275]); ACMG classification with functional support for VUS ([PMID: 42057034]).
- **Histopathology:** Streak gonads; surveillance for gonadoblastoma/dysgerminoma.
- **Differential diagnosis:** NR5A1, SRY, SOX9, MAP3K1, WT1, GATA4, AR (androgen insensitivity), HSD17B3, and 45,X/46,XY mosaicism ([PMID: 42365275], [PMID: 42057034]).

11. Outcome / Prognosis

- **Survival:** Life expectancy is essentially normal with appropriate management; the major life-threatening risk is malignant transformation of retained dysgenetic gonads.
- **Morbidity:** Infertility, need for lifelong hormone replacement, and cancer risk (~15–23% in dysgenesis; [PMID: 27862157]).
- **Prognostic factors:** Degree of gonadal dysgenesis (dysgenesis > regression for tumor risk); familial cases may carry higher tumor risk ([PMID: 38337479]).

12. Treatment

- **Hormone replacement therapy** (estrogen ± progestin, or testosterone as appropriate) — NCIT: Hormone Replacement Therapy.
- **Prophylactic/therapeutic gonadectomy** of dysgenetic Y-bearing gonads — NCIT: Gonadectomy/Orchiectomy.
- **Fertility counseling** (typically infertile; assisted reproduction/oocyte donation for uterus-bearing individuals).
- **Psychosocial and gender-affirming care**, individualized.
- **Genetic counseling** (AD recurrence, variable expressivity/penetrance, distinct NEDBAVC risk).
- No gene/RNA/cell therapy, disease-specific drug, or pharmacogenomic guidance exists.

13. Prevention

- **Primary:** Not applicable (congenital genetic condition).
- **Secondary:** Cascade genetic testing of at-risk relatives; early identification enabling timely tumor surveillance and gonadal management.
- **Tertiary:** Prophylactic gonadectomy to prevent malignancy; hormone replacement to prevent osteoporosis and secondary-sex-characteristic deficits.
- **Reproductive options:** Genetic counseling, prenatal/preimplantation genetic testing for known familial variants.

14. Other Species / Natural Disease

- **Orthologs:** Mouse *Dhx37* (used in the conditional-knockout model — [PMID: 41535247]); yeast ortholog *DHR1* (structural/functional studies — [PMID: 31188444]). NCBI Gene (human) 57647.

- **Natural disease:** No naturally occurring companion-animal or wildlife DHX37 sex-reversal disease is documented in OMIA to date.
- **Conservation:** The ribosome-biogenesis function of DHX37/Dhr1 is deeply conserved from yeast to human; the sex-determination consequence is a vertebrate/mammalian-specific downstream effect.

15. Model Organisms

- **Mouse:** Cell-specific (supporting-cell) conditional *Dhx37* knockout recapitulates impaired testis development via nucleolar stress/p53 apoptosis ([PMID: 41535247]) — the best available model; limitation: it is a knockout, not a patient-allele knock-in, so dominant/hotspot-allele effects are not directly modeled.
 - **Yeast:** Dhr1 structural/functional studies define the helicase mechanism (U3 snoRNA displacement, C-terminal domain) and the biogenesis consequences of DHX37 mutations ([PMID: 31188444]); limitation: no sex-development readout.
 - **Gaps:** No zebrafish/invertebrate DHX37 sex-development model; no humanized R308Q/R674W knock-in. Resources: MGI (mouse), SGD (yeast).
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Evidence Base

PMID	Title (abbrev.)	Evidence type	Role in report
31337883	DHX37 variants a frequent cause of 46,XY GD/TRS	Human cohort (n=145)	Establishes DHX37 as SRXY11 gene, AD inheritance, spectrum (F1)
41535247	Multi-omics of Dhx37 deficiency on testis/nucleolar homeostasis	Mouse KO, in vitro	Core mechanism: nucleolus/PI3K-AKT/p53/apoptosis (F2)
35835064	DHX37 and 46,XY DSD: A new ribosomopathy?	Review	Allelic NEDBAVC vs non-syndromic DSD; ribosomopathy framing (F3)
27862157	Gonadal tumour risk in 292 phenotypic females with Y material	Human cohort (n=292)	Quantifies GCT risk; supports gonadectomy (F4)
34293745	Expanding DSD phenotypes with DHX37 variants	Human clinical	Spectrum poles; variable expressivity; mechanism unknown (F5, F8)
42510869	Novel and known DHX37 variants	Human clinical	Confirms mild-end heterogeneity (F5)
37497464	Late presentation of Swyer syndrome	Case report	Hypergonadotropic hypogonadism signature (F6)
41466375	Variations in sex characteristics across OMIM	Analysis	Supports genome-wide sequencing in DSD (F6)
31287541	DHX37 defects and 46,XY GD spectrum	Human cohort (n=87), IHC	Recurrent hotspots, gnomAD enrichment, ETRS, expression (F7)
31188444	Dhr1 C-terminal domain essential for SSU biogenesis	Structural/yeast	Molecular basis of helicase role; U3 snoRNA displacement
38337479	Gonadoblastoma in familial Swyer syndrome	Case + review	Familial tumor-risk context

PMID	Title (abbrev.)	Evidence type	Role in report
31883875 / 32057790	Tumor risk in 45,X/46,XY mosaicism	Human clinical	Differential diagnosis / tumor-risk context
42365275 / 42057034 / 40916030 / 39829003	WES cohorts identifying DHX37 (with NR5A1 etc.)	Human cohorts	Diagnostic yield, VUS interpretation, differential genes
42151440	DHX37 in breast/ovarian cancer prognosis	Human tumor cohorts	Non-DSD context: DHX37 tissue-context-dependent roles

Where the evidence converges: Independent human cohorts ([31337883], [31287541]) establish DHX37 causation, hotspot recurrence, and statistical enrichment; the mouse multi-omics study ([41535247]) supplies the cell-and-pathway mechanism (Sertoli apoptosis via nucleolar stress/p53/PI3K-AKT); tumor-risk data ([27862157]) drive the management recommendation.

Where the evidence is in tension / incomplete: Human IHC localizes DHX37 mainly to germ and Leydig cells, "rarely" to Sertoli cells ([31287541]), whereas the mechanistic mouse model centers on Sertoli-cell apoptosis ([41535247]) — leaving open whether the human phenotype arises cell-autonomously in supporting cells, non-cell-autonomously from germ/Leydig-cell dysfunction, or both. Reviews explicitly state the mechanism of tissue specificity is unknown ([34293745]).

Limitations and Knowledge Gaps

- 1. Unresolved tissue-specificity paradox (central gap).** No experimental demonstration explains why a ubiquitous 40S-biogenesis factor selectively disables gonadal determination ([PMID: 34293745]).
- 2. Allele-specific functional data are sparse.** The dominant-negative vs hypomorphic nature of hotspot alleles (R308Q, R674W) is inferred, not proven; some clinically encountered variants remain VUS ([PMID: 42057034]).

3. **Cell-of-origin ambiguity.** Human expression (germ/Leydig-dominant) versus mouse phenotype (Sertoli apoptosis) are not fully reconciled.
 4. **No humanized/knock-in model** of the recurrent DSD alleles, and **no zebrafish/invertebrate sex-development model** exists; the mouse model is a conditional knockout.
 5. **Incomplete penetrance/variable expressivity** (e.g., fertile carrier fathers) implies unidentified genetic/environmental modifiers; none are established.
 6. **Epidemiology is imprecise** — DHX37-specific prevalence/incidence are unquantified; data come from enriched DSD cohorts, not population registries.
 7. **No environmental/infectious contributors** are known; environmental sections are largely not applicable.
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Proposed Follow-up Experiments / Actions

1. **Generate patient-allele knock-in mice** (Dhx37^{R308Q/+}, R674W^{+/+}) to test dominant-negative behavior and recapitulate the graded dysgenesis→regression phenotype in vivo.
 2. **Cell-type-resolved fetal gonad profiling** (single-cell/spatial transcriptomics of human and mouse fetal gonad) to determine whether the primary lesion is in Sertoli, germ, or Leydig cells and to map ribosome-flux dependence during the determination window. (Search: Human Cell Atlas, GEO, CELLxGENE.)
 3. **Allele-specific functional assays** — 40S maturation, U3 snoRNA displacement, nucleolar-stress and p53 readouts — for each recurrent and VUS allele to move variants from VUS toward pathogenic/benign per ACMG.
 4. **Zebrafish dhx37 sex-development model** to exploit tractable gonad genetics and test conservation of the mechanism.
 5. **Prospective natural-history/registry study** capturing penetrance, tumor incidence and timing, and modifier genotypes across DHX37 carriers, to refine gonadectomy timing and counseling.
 6. **Test the tissue-specificity hypotheses directly** — e.g., titrate p53 activity (Trp53 co-deletion) in the Dhx37 model to establish whether p53-driven apoptosis is necessary and sufficient for the gonadal phenotype.
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Conclusion

46,XY Sex Reversal 11 (SRXY11; OMIM #273250) is an autosomal-dominant disorder of sex development caused by heterozygous, largely recurrent missense variants in the ribosome-biogenesis helicase *DHX37*. It presents as a phenotypic spectrum from complete gonadal dysgenesis (Swyer) to testicular regression/anorchia and mild male undervirilization, and is best understood as a putative ribosomopathy in which nucleolar stress, p53-driven apoptosis, and reduced PI3K-AKT survival signaling cause failure/regression of the fetal testis-supporting lineage. Diagnosis combines a discordant 46,XY karyotype, hypergonadotropic hypogonadism, and molecular sequencing; management is supportive — hormone replacement, prophylactic gonadectomy of Y-bearing dysgenetic gonads (~15–23% tumor risk), and genetic counseling. The defining open question is how a housekeeping ribosome factor produces a gonad-restricted phenotype.

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